The splittable flow formulation

Lab handout 3

October 2026

Files: formulations/splittable_flow.py and labs/lab1_first_ilp.py.

1. The problem

A network is given, with a capacity (number of wavelengths) on each link, together with a traffic matrix whose demands are also expressed in wavelengths. The problem is to route all the demands using the minimum number of wavelengths, summed over all links. For instance, a demand of 5 wavelengths routed on a path of 3 links contributes 15 to the objective. In this first formulation the traffic of a demand may be split over several paths, for example 10 wavelengths on one path and 4 on another.

Link-flow formulation

The notation and its counterpart in the code are summarised in the following table.

symbol	meaning	code
$G = (N, E)$	directed network, $e = (i, j)$ a link	`G`, a `networkx.DiGraph`; `for (i, j) in G.edges`
u_e	capacity of link e (wavelengths): f_e fibers of W_e wavelengths, $u_e = f_e\, W_e$	`G.edges[i, j]["fibers"] * G.edges[i, j]["capacity"]`
D	set of demands; s_d, t_d, h_d origin, destination and wavelengths of d	`demands`; `d.origin`, `d.destination`, `d.traffic`
$\delta^+(n)$, $\delta^-(n)$	links leaving and entering node n	`G.out_edges(n)`, `G.in_edges(n)`
$x_{d,e}$	wavelengths of demand d carried by link e (non-negative integer variables)	`x[d.id, i, j]`

$$
\begin{aligned}
&\textit{minimise} && \Sigma_{d \in D} \Sigma_{e \in E}\, x_{d,e} \\
&\textit{subject to} && \Sigma_{e \in \delta^+(n)}\, x_{d,e} - \Sigma_{e \in \delta^-(n)}\, x_{d,e} = h_d \text{ if } n = s_d,\ -h_d \text{ if } n = t_d,\ 0 \text{ otherwise}, \quad \text{for all } d \in D,\ n \in N \\
& && \Sigma_{d \in D}\, x_{d,e} \le u_e, \quad \text{for all } e \in E \\
& && x_{d,e} \ge 0 \text{ and integer}
\end{aligned}
$$

The first group of constraints expresses flow conservation: h_d wavelengths leave the origin of demand d, the same amount reaches its destination, and at every other node the flow entering is equal to the flow leaving. The second group limits the load of each link to its capacity. The variables are integer, since they count wavelengths. For Spain-7 (16 links and 7 nodes) with 24 demands the model has $24 \times 16 = 384$ variables and $24 \times 7 + 16 = 184$ constraints.

The code

The file `formulations/splittable_flow.py` is short, and each block corresponds to one part of the model. The variables are created for every demand and every link:

```
x = {}
for d in demands:
    for (i, j) in G.edges:
        x[d.id, i, j] = prob.add_variable(f"x_{d.id}_{i}_{j}", lowBound=0, cat=pulp.LpInteger)
```

The objective function and the capacity constraints are

```
prob += pulp.lpSum(x[d.id, i, j] for d in demands for (i, j) in G.edges)
```

```
for (i, j) in G.edges:
    u = G.edges[i, j]["fibers"] * G.edges[i, j]["capacity"]     # fibers x wavelengths per fiber
    prob += pulp.lpSum(x[d.id, i, j] for d in demands) <= u, f"capacity_{i}_{j}"
```

The flow conservation constraints are reproduced at the end of handout 2. The function `solve_ilp` then calls the solver (CBC or HiGHS), and the flow of each demand is decomposed into paths by `flow_to_routes`. The resulting `Solution` is checked, saved and displayed by PyNet2Plan.

2. The data

The network is Spain-7 (handout 2), with 8 pairs of links, one fiber per link and the same capacity C on every link (40 wavelengths in the topology file). The traffic matrix is `data/traffic/spain7_traffic_split_vs_unsplit.csv`: 24 demands of 4 to 16 wavelengths, 229 wavelengths in total. The demands are few and large, and several of them compete for the same links.

Two values help to interpret the results. The first is a lower bound on the objective. If the capacities were unlimited, each demand would follow a path with the fewest hops, and the objective would be the sum of h_d times the number of hops of that path, which is 408. No routing can use fewer than 408 wavelengths. The second is the cut bound of handout 2: 68 wavelengths must reach Malaga and Murcia through only two links, so the capacity must be at least 34.

3. Running the formulation

In the graphical interface:

1. open the network and its traffic with *File > Open example > Spain-7: splittable vs unsplittable* (all links have C = 40);
2. select *Splittable flow formulation* in the run bar and press *Run*;
3. in the summary press *Open in the Results tab* and examine the *Report*, *Routes*, *Links* and *Occupancy* tabs (sections 11 to 14 of handout 1).

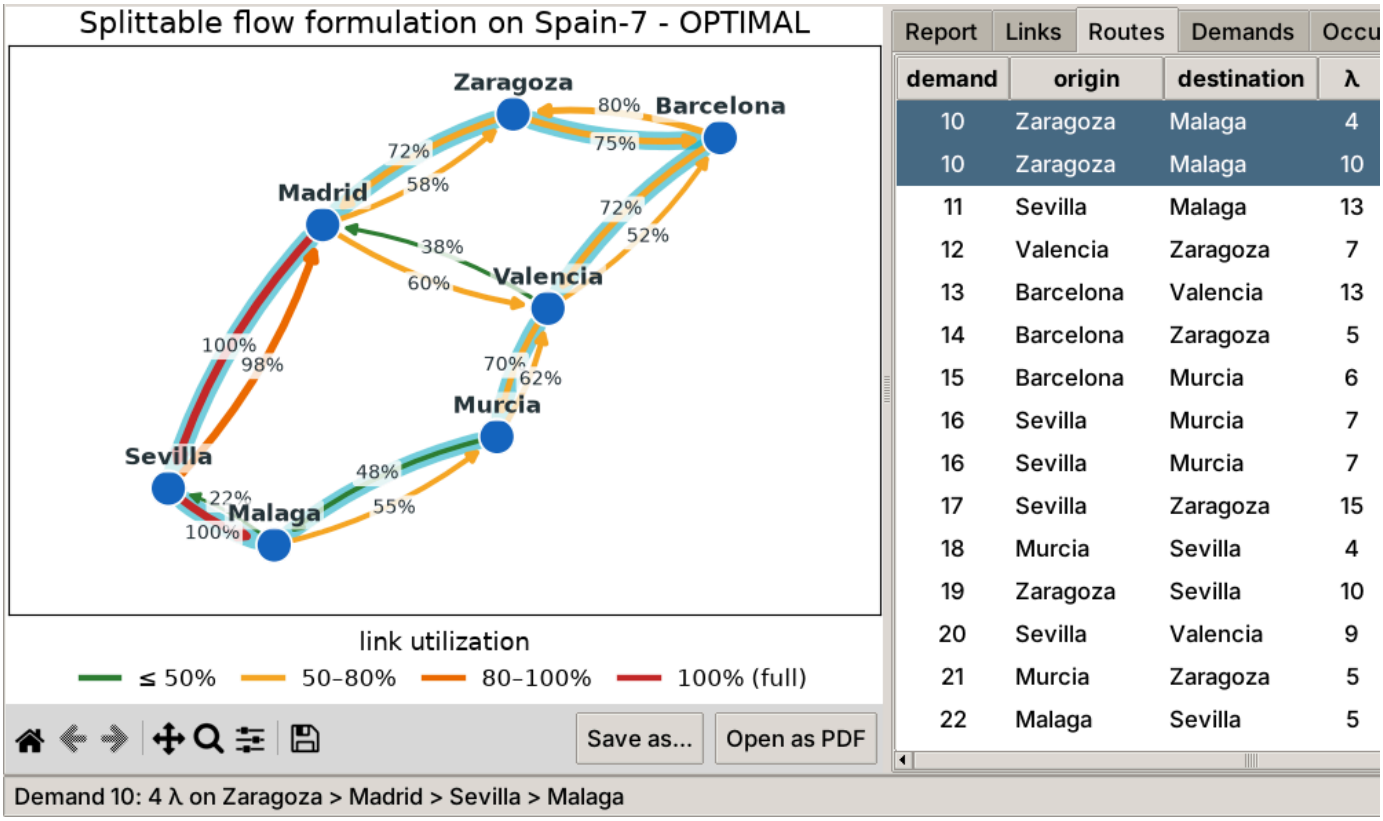


Figure 1. Result for C = 40. The paths highlighted in light blue are the two routes of demand 10 (Zaragoza to Malaga).

The first part of the script `labs/lab1_first_ilp.py` performs the same run from the command line and prints some additional information:

```
python labs/lab1_first_ilp.py
```

```
status OPTIMAL, objective 425.0, time 0.031 s
objective - lower bound = 425 - 408 = 17 extra wavelengths (detours)
saturated links: Madrid->Sevilla, Sevilla->Malaga
  demand 10 Zaragoza->Malaga (14 λ, fewest hops 3):  4 λ on Zaragoza > Madrid > Sevilla > Malaga;  10
λ on Zaragoza > Barcelona > Valencia > Murcia > Malaga   -> 10 extra
  demand 16 Sevilla->Murcia (14 λ, fewest hops 2):  7 λ on Sevilla > Malaga > Murcia;  7 λ on Sevilla
> Madrid > Valencia > Murcia   -> 7 extra
  total extra = 17  (= objective - lower bound)
saved in results/2026-10-26_17-22-12_splittable-flow_spain-7
figure: results/lab1_first_ilp/ilp_map_c40.png
figure: results/lab1_first_ilp/ilp_map_c40.pdf
```

Discussion

The status is OPTIMAL and the objective is 425, that is 17 wavelengths more than the lower bound. Each unit above 408 corresponds to one wavelength carried on a link that is not part of a fewest-hop path, i.e. to a detour; the extra wavelengths listed in the output add up to 17. Detours are necessary because some links are full: the demands that would use them have to be routed around them. The full links can be identified on the map and in the *Links* tab, where a click on a link lists the demands that use it. Finally, some demands are split over two paths. The formulation allows it, and splitting allows the solver to fill the scarce links exactly.

In general several routings use exactly 425 wavelengths, and the solver returns one of them. A different solver, or another version of CBC, may return a different optimal solution, with other split demands and

other full links. The optimal value, 425, does not change.

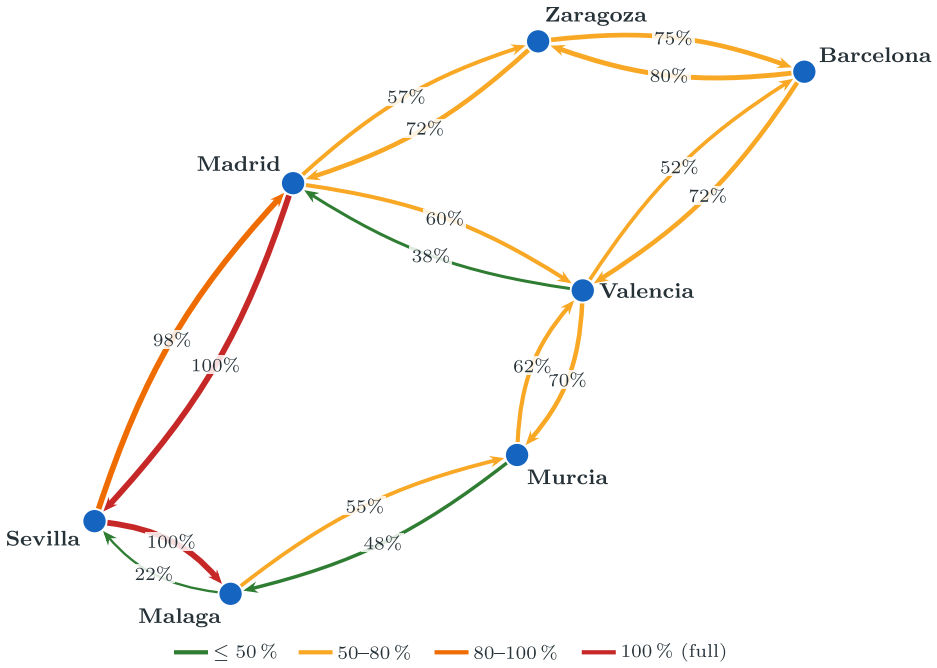


Figure 2. Link utilization in the optimal solution for C = 40 (same colours as in the graphical interface).

4. Experiment 1: the effect of the capacity

The formulation is now solved for every capacity from C = 30 to C = 60, either with *Compare (sweep)…* in the graphical interface (capacities `30-60`, see sections 16 to 18 of handout 1) or with the second part of the script:

C	status	objective	split	full links	time ms
30	INFEASIBLE	-	-	-	24.1 ± 3.4
31	INFEASIBLE	-	-	-	22.6 ± 5.4
32	INFEASIBLE	-	-	-	22.0 ± 3.3
33	INFEASIBLE	-	-	-	22.9 ± 3.5
34	OPTIMAL	431.0	2	4	30.5 ± 3.4
35	OPTIMAL	430.0	2	2	39.9 ± 26.2
36	OPTIMAL	429.0	2	2	37.0 ± 18.0
37	OPTIMAL	428.0	2	2	37.6 ± 15.6
38	OPTIMAL	427.0	2	2	30.4 ± 3.6
39	OPTIMAL	426.0	2	3	31.6 ± 7.0
40	OPTIMAL	425.0	2	2	30.2 ± 2.2
41	OPTIMAL	424.0	2	2	28.8 ± 1.8
42	OPTIMAL	423.0	2	2	27.6 ± 1.3
43	OPTIMAL	422.0	2	2	27.3 ± 1.5
44	OPTIMAL	421.0	2	2	29.7 ± 4.5
45	OPTIMAL	420.0	2	2	29.7 ± 3.8
46	OPTIMAL	419.0	2	2	30.2 ± 4.2
47	OPTIMAL	418.0	2	2	28.9 ± 1.6
48	OPTIMAL	417.0	2	2	29.3 ± 3.6
49	OPTIMAL	416.0	2	2	30.0 ± 3.5
50	OPTIMAL	415.0	1	2	28.8 ± 2.6
51	OPTIMAL	414.0	1	1	27.2 ± 2.3
52	OPTIMAL	413.0	1	1	27.9 ± 2.1

```
  53     OPTIMAL     412.0     1          1       27.0 ± 1.6
  54     OPTIMAL     411.0     1          1       27.9 ± 4.1
  55     OPTIMAL     410.0     1          1       28.6 ± 3.0
  56     OPTIMAL     409.0     1          1       30.6 ± 5.5
  57     OPTIMAL     408.0     0          1       28.9 ± 2.6
  58     OPTIMAL     408.0     0          0       31.8 ± 8.0
  59     OPTIMAL     408.0     0          0       33.8 ± 9.9
  60     OPTIMAL     408.0     0          0       28.9 ± 2.7
time: mean ± standard deviation of 5 runs
smallest feasible capacity: C = 34;  the lower bound 408 is reached from C = 57
sweep saved in results/2026-10-26_17-22-13_sweep_spain-7
figure: results/lab1_first_ilp/ilp_objective_vs_capacity.png
figure: results/lab1_first_ilp/ilp_objective_vs_capacity.pdf
figure: results/lab1_first_ilp/ilp_split_and_full_links.png
figure: results/lab1_first_ilp/ilp_split_and_full_links.pdf
```

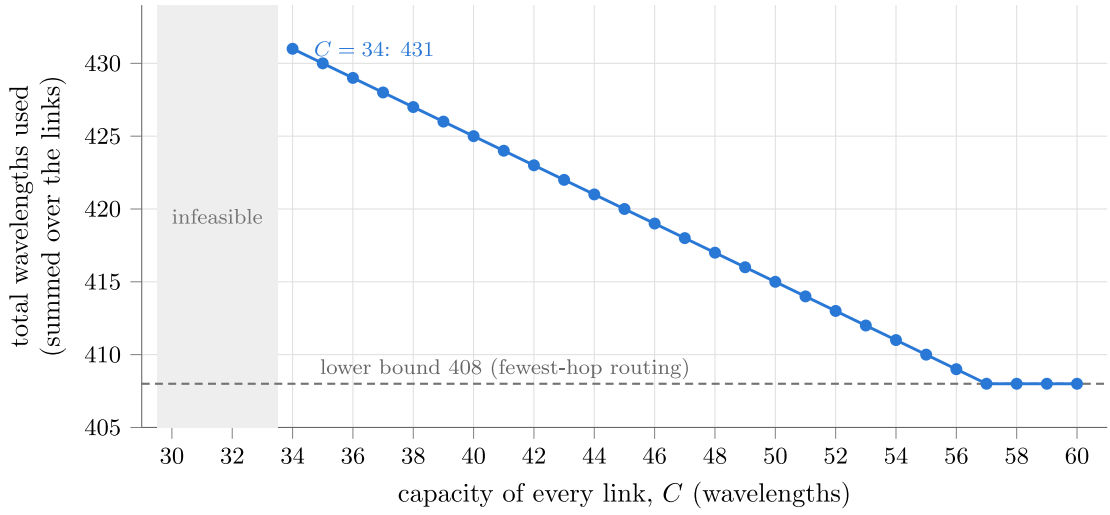


Figure 3. Objective as a function of the capacity. The shaded region contains the infeasible capacities.

Three regions can be distinguished. For C ≤ 33 the problem is infeasible, as predicted by the cut bound of handout 2, which is tight in this case. Between C = 34 and C = 57 the objective decreases by exactly one wavelength for each additional wavelength of capacity: one more wavelength on the bottleneck links allows one more wavelength of some demand to avoid a detour of one hop. For C ≥ 57 the objective equals the lower bound 408, every demand follows a fewest-hop path and additional capacity is not used.

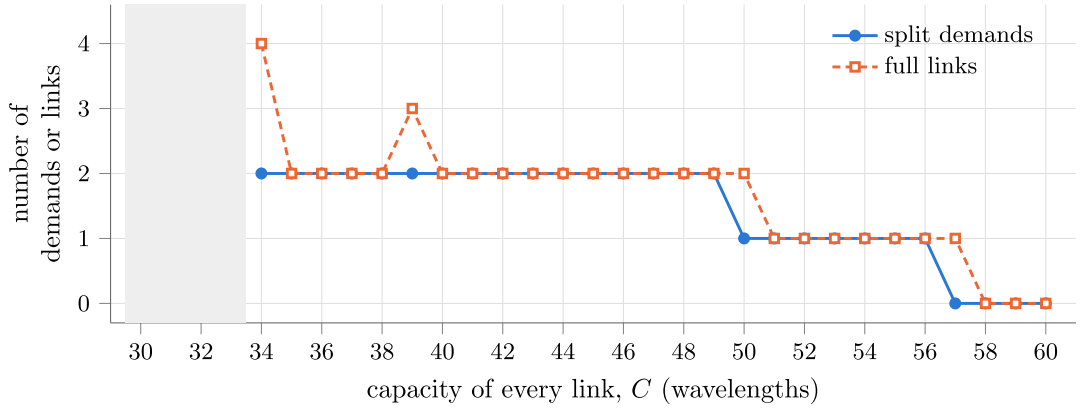


Figure 4. Number of split demands and of full links in the optimal solution returned by the solver. These numbers depend on which optimal solution is returned.

5. Experiment 2: fibre cuts

A fibre cut interrupts both directions of a link. In the graphical interface the two links are selected in the *Links* tab (Ctrl+click, or Cmd+click on macOS) and switched off with *Fail / repair selected*; the capacity is set with *Same capacity for all…*. The third part of the script tries the 8 possible cuts with C = 60:

```
no failure: OPTIMAL 408.0
  cut Madrid-Zaragoza            OPTIMAL  472.0
  cut Madrid-Sevilla          INFEASIBLE
  cut Madrid-Valencia            OPTIMAL  469.0
  cut Barcelona-Zaragoza         OPTIMAL  442.0
  cut Barcelona-Valencia         OPTIMAL  455.0
  cut Valencia-Murcia         INFEASIBLE
  cut Sevilla-Malaga          INFEASIBLE
  cut Malaga-Murcia           INFEASIBLE
figure: results/lab1_first_ilp/ilp_fiber_cuts_c60.png
figure: results/lab1_first_ilp/ilp_fiber_cuts_c60.pdf

figures saved in results/lab1_first_ilp
```

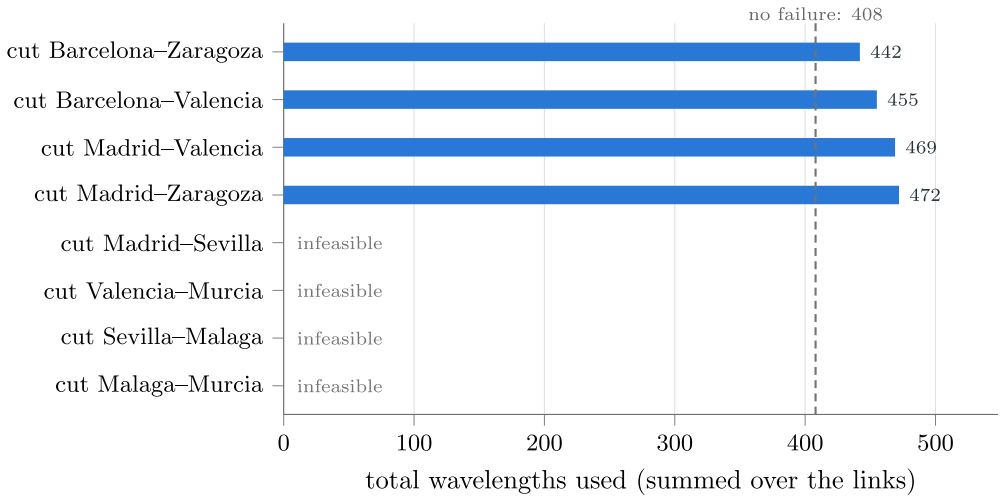


Figure 5. The 8 single fibre cuts with C = 60 on every link.

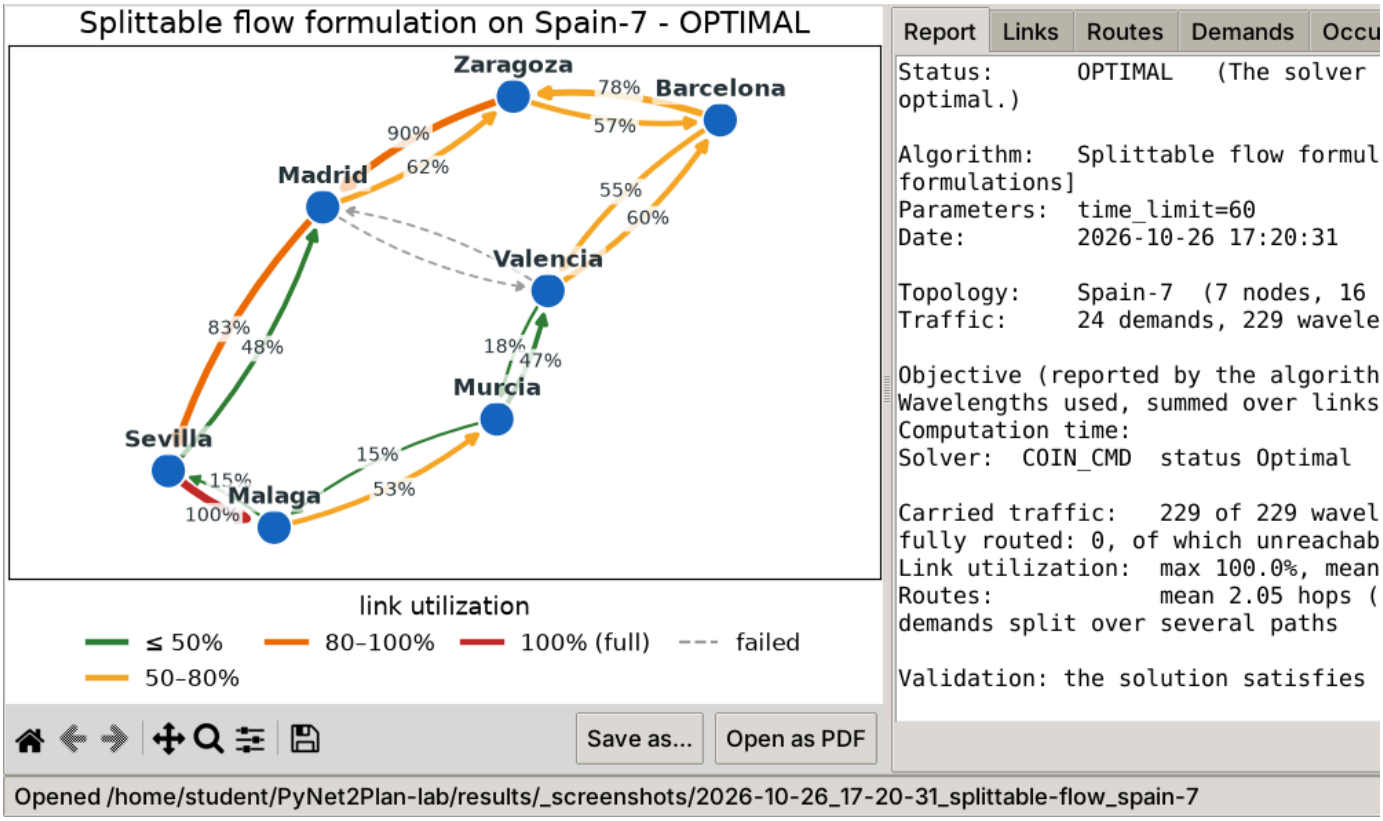


Figure 6. C = 60 with the link Madrid-Valencia cut (dashed): the traffic is carried by the remaining links, with longer detours.

With C = 60 the network survives four of the eight cuts, at the cost of up to 64 additional wavelengths, while the other four cuts make the problem infeasible. A capacity that is sufficient in normal operation does not make a network survivable.

6. Charts for the lab report

The figures of this handout and those saved by the script in `results/lab1_first_ilp/` (PNG for the screen, PDF for a report, with their data in JSON files) follow a few rules, which also apply to the charts of the lab reports.

- Every axis has a title with the quantity and its unit, for example *capacity of every link, C (wavelengths)* or *computation time (ms)*.
- Tick labels, axis titles and legends remain readable when the figure is reduced to the width of the page: in the final document they should not be smaller than about 8 points.
- The series differ in colour, marker and line style and are named in a legend or directly on the chart, so that the chart can also be read in a black and white print and by readers with a colour vision deficiency. Hollow markers keep a series visible where it overlaps another one.
- Missing points are shown explicitly, for example with a shaded band or with crosses for the capacities without a solution.
- Integer quantities, such as capacities and wavelengths, have integer ticks.
- A quantity that changes from one measurement to the next, such as a computation time, is measured several times. The chart shows the mean with error bars, and the caption states what the error bars represent (in this course, one standard deviation).
- Charts are inserted in a report as vector files (PDF), which remain sharp at any size. The buttons *Save as...* and *Open as PDF* of the graphical interface produce such files.

In Matplotlib, `plotting.use_report_style()` and `plotting.series_style(i)` of PyNet2Plan apply these settings; their use can be seen in the functions at the end of the lab scripts.

7. Questions

The questions can be answered with the graphical interface or by modifying the script. For every result, record the network, the traffic file, the capacity and the parameters used.

1. For C = 40, list the full links and the demands that do not follow a fewest-hop path, and check that the additional wavelengths add up to 425 − 408. Is your solution the same as the one obtained by other students?
2. Explain with the cut bound why the problem is infeasible for C = 33. Which links are full for C = 34?
3. What is the smallest capacity, equal on all links, for which the network survives every single fibre cut? The option `--failure-capacity` of the script can be used, with values such as 64, 66, 68.
4. Starting from C = 34, capacity can be added to individual links instead of to all of them. Which links would you upgrade, and by how much, to reach the lower bound 408 with the smallest total amount of additional capacity?
5. Look at the computation times of the sweep, given as the mean and the standard deviation of 5 runs. Are the differences between the capacities larger than the standard deviations? Repeat the run on a larger network (*File > Open example > NSFNET-14*) and compare the number of variables and the computation time.

8. Next lab

In the next lab splitting is no longer allowed: in `formulations/unsplittable_flow.py`, which is downloaded with `git pull` before the lab (handout 0), each demand must use a single path. The formulation has to be completed and then compared with the splittable one on the same data. The traffic matrix of this handout was chosen so that the two formulations give different results for most capacities.